

CoRoT-2b

R-0134

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_250718 · created 2026-07-16T06:34:54+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460874.8645 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.163 +/- 0.018
Transit depth (Rp/Rs)^2	2.66 +/- 0.6 [%]
Orbital Inclination (inc)	89.59 +/- 1.5
Airmass coefficient 1 (a1)	1.062 +/- 0.024
Airmass coefficient 2 (a2)	-0.039 +/- 0.012
Scatter in the residuals of the lightcurve fit is	2.25 %
Best Comparison Star	#2 - [234, 280]
Optimal Method	PSF photometry
Transit Duration (day)	0.0956 +/- 0.0024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.163 ± 0.018 vs 0.1667 (deviation 0.21σ)
Duration fit vs published	0.0956 d vs 0.095 d (deviation 0.25σ)
Mid-time O-C	-6.5 min
Depth SNR	9.1
Residual scatter	2.25 %

Light curve

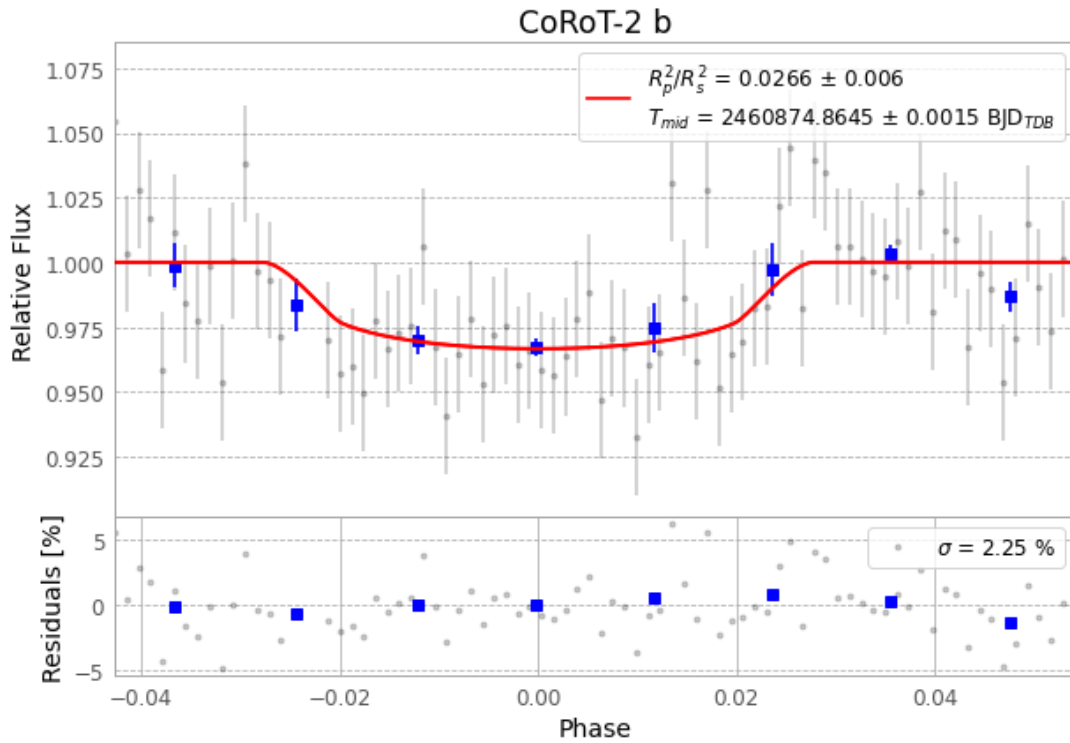


Figure 1 — FinalLightCurve_CoRoT-2 b_2025-07-17.png (SHA-256 71c07a5362efb6e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 253], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e18066a26bdea6e1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

82 FITS frames (54 MB), CoRoT-2250718064816.FITS → CoRoT-2250718105116.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-250718.log (2781f81aaf35...), FinalLightCurve_CoRoT-2 b_2025-07-17.png (71c07a5362ef...), FinalLightCurve_CoRoT-2 b_2025-07-17.csv (13cb9c51285c...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 06:34Z.